When More Observation Hurts: Redundancy and Robust Clarification in Ask-or-Act Under Instruction Ambiguity

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Abstract. In cooperative assistance an agent receives goal evidence through two channels: passive inference from observed actions and active clarification via costly questions. Using a controlled gridworld benchmark we show that these channels are *strongly redundant* when passive inference is reliable, but redundancy weakens as passive quality degrades—making clarification valuable primarily as robustness insurance. A Shapley-style decomposition reveals an interaction term of −38 pp at K=3 (strong redundancy) that shrinks to −17 pp when 60% of actions are unobserved. We further show that assistability is non-monotonic in principal rationality: a *more* rational principal can be harder to assist because action aliasing hides intent. Under model mismatch, additional observation actively poisons the posterior (−2.2 pp/step vs. −0.8 matched), explaining why myopic passive-aware heuristics fail to improve over a simple cost-comparing baseline.

1 Introduction

A central challenge in cooperative assistance is deciding when a helper should ask for clarification rather than act. Acting immediately risks committing to the wrong goal, whereas asking too often wastes time and may frustrate the principal. The trade-off depends on the informativeness of available questions, the cost of a wrong action, and what the principal's behaviour already reveals.

We study this question by constructing an attribute-structured, post-parsing clarification benchmark: the candidate goal set is known, question semantics are fixed, and the principal model is given. The benchmark's value is controlled diagnosis, not ecological realism. It includes a cooperative principal whose actions are observable for passive inference via inverse planning, and a question bank the assistant may use at explicit cost for active clarification. The contributions are threefold: a reproducible two-room gridworld with templated instruction ambiguity and a Boltzmann principal model, a systematic comparison of six ask-or-act policies across 19,800 episodes, and an empirical characterisation of how the two channels interact. Our contribution is not a new ask policy, but, to our knowledge, the first controlled characterisation of how the value of active clarification changes as passive action evidence weakens, with the empirical success–question-cost frontier shifting substantially with principal rationality.

Why a controlled benchmark? Prior work studies each channel in isolation: CLIPS [15] does inverse planning without asking; ATD [11], SAGE [14], and Grand et al. [5] study clarification without action-based inference. To our knowledge, no prior benchmark combines both channels with controlled ambiguity sweeps (Table 1).

The gap is the ability to study how passive and active evidence *interact*—substituting, complementing, or even conflicting—under a shared objective.

Three Takeaways.

1. **Passive and active channels are strongly redundant, not additive.** A Shapley-style decomposition shows an interaction of −38 pp at K=3. Redundancy weakens as passive quality degrades (−38 → −17 pp), so clarification is most valuable as robustness insurance.
2. **Assistability is non-monotonic in principal rationality.** At K=3, NeverAsk success peaks at intermediate β and drops at high β: extreme rationality collapses action diversity and hides intent.
3. **Under model mismatch, more observation actively poisons the posterior.** Forced observation costs −2.2 pp/step under mismatch vs. −0.8 matched (2.8× worse), explaining why passive-aware heuristics cannot outperform the simple cost-comparing baseline.

Table 1: Illustrative comparison to closest prior work (not an exhaustive survey). ✓ = present; ∘ = partial; – = absent.

Feature	CLIPS	GRD	PAGR	ATD	ShootFirst	**Ours**
Action-based goal inference	✓	∘	✓	–	–	✓
Explicit ask-or-act decision	–	–	∘	✓	✓	✓
Typed questions with cost c_q	–	–	–	✓	–	✓
Task completion (not just recog.)	✓	–	–	✓	✓	✓
Controlled ambiguity-K sweep	–	–	–	–	–	✓
Ground-truth task success	∘	–	–	–	∘	✓

Beyond the present comparison, the benchmark is intended as a reusable diagnostic testbed for future ask-or-act methods, enabling controlled evaluation as ambiguity, action informativeness, and question cost vary independently.

2 Related Work

Goal inference and inverse planning. Inverse planning infers goals from behaviour [1, 12]; CIRL [6] formalises cooperative teams with reward uncertainty; CLIPS [15] combines action and language evidence. None provides an explicit ask-or-act decision.

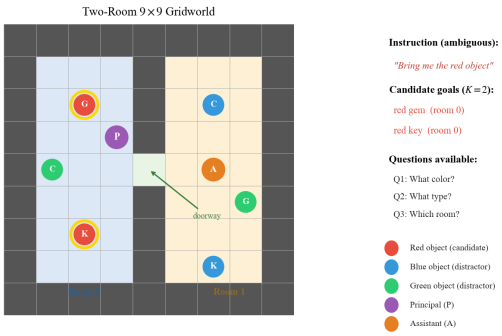

Figure 1: Two-room 9×9 gridworld layout. Room 0 (blue, left) and Room 1 (yellow, right) are separated by a wall with a single doorway. Gold rings mark the K=2 candidate goals that match the instruction "Bring me the red object". P = principal, A = assistant.

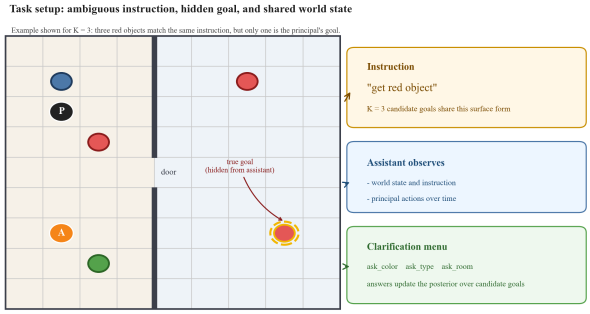

Figure 2: Task setup. The principal has a hidden goal in a two-room gridworld; the assistant sees the world state, an ambiguous instruction, and the principal's movement. Questions can collapse the posterior over candidate goals.

Shared autonomy and HRI clarification. Javdani et al. [8] blend assistance via hindsight optimisation; Tellex et al. [16] and Cakmak and Thomaz [3] study robot clarification questions. Our setting decouples the principal and assistant and adds an explicit question cost.

Goal recognition design and active sensing. GRD [10] reduces shared action prefixes to improve legibility; Dragan et al. [4] study legible motion; Zhang et al. [17] combine passive recognition with active information gathering in grids. Our work differs in studying *task completion* with typed costly questions and a shared success/cost objective.

Ask-or-act under explicit cost. ATD [11] and SAGE [14] frame LLM clarification as BED/EVPI; Grand et al. [5] study ask-vs-act in language games. None includes a cooperative principal whose actions are observed for inverse planning, precluding measurement of channel interaction.

3 Method

3.1 Environment

We use a 9×9 two-room gridworld with M=6 object instances per episode. A vertical wall at the grid midpoint divides the space into *Room 0* on the left and *Room 1* on the right, connected by a single doorway at the central cell (Figure 1). Each episode samples 6 objects from the type set $\{\text{key}, \text{gem}, \text{coin}\}$ and colour set $\{\text{red}, \text{blue}, \text{green}\}$; exactly K match the ambiguous instruction (candidate goals) and the remaining $M-K$ are distractors.

A principal has a hidden goal object. An instruction template maps the goal to a surface form shared by K candidate goals (ambiguity level K). The principal takes approximately rational actions towards its goal, sampled from a Boltzmann policy with rationality β and ε-greedy noise: $\pi_{\beta,\varepsilon}(a \mid s, g) = (1-\varepsilon)\,\text{softmax}_\beta\big(V(s,\cdot,g)\big)_a + \varepsilon/|A|$, where $V(s,a,g) = -d^*(\text{succ}(s,a), g)$ is the negated shortest-path distance from the successor of a to g.

The assistant observes the state, instruction, and all principal actions, but not the true goal. An episode ends when the assistant picks an object (success if correct) or when the step budget $T_{\max}$=200 is exhausted. The question bank Q contains three types of questions addressing the goal's colour, room, and object type, each with a factual answer drawn from a per-type noisy channel. When the assistant decides to ask, it remains at its current position. When it decides to act, it takes one step toward the MAP-goal object via A* navigation and then re-evaluates.

Running example. Consider the instruction "bring me the red object" with K=3 candidates (red key, red gem, red coin). At step 1, the posterior is uniform at $\frac{1}{3}$ each. *AskOrAct* evaluates $\text{CostAct}(b_t) = \mathbb{E}_g[\ell(s_t, g)]$ against $\text{CostAsk}(q^*) = 1 + c_q + \mathbb{E}_\alpha[\text{CostAct}(b_{t,q,\alpha})]$. With c_q=0.5, the posterior variance is high enough that asking "what type?" yields a lower expected cost, and the agent asks. After receiving the answer "key", the posterior concentrates on the red key and the agent acts directly. *InfoGainAsk* reaches the same first-step decision but does not include the explicit cost comparison. It asks on every step where entropy exceeds the gate, selecting the highest information-gain question regardless of cost efficiency, which typically leads to more questions overall. Table 2 summarises all hyperparameters. Figure 2 shows the full cooperative setup.

Table 2: Environment and policy hyperparameters.

Parameter	Value
Grid size	9×9, two rooms
Objects	6 (3 types × 3 colours)
Step budget $T_{\max}$	200
Question bank $\lvert Q\rvert$	3 (color, room, object)
Ask budget $Q_{\max}$	3
Ask window	steps ≤ 6
Question cost c_q	0.5
Entropy gate τ_H (AskOrAct)	0.3
IG threshold τ_{IG}	0.01
Answer noise (color/room/object)	0.05 / 0.10 / 0.20
Principal rationality β	2.0 (default)
Principal noise ε	0.05 (default)

3.2 Bayesian Posterior

Per-step protocol. At each step: (1) observe the principal's action, (2) update b_t from the action, (3) choose ask or act, (4) if ask, receive answer and update b_t; if act, move one step via A* toward the MAP goal, then repeat.

The assistant maintains a posterior $b_t(g)$ over candidate goals $\mathcal{G}$. Initialisation uses the instruction likelihood under each goal. After observing principal action a_t at state s_t:

$$b_{t+1}(g) \propto b_t(g) \cdot \pi_{\hat{\beta},\hat{\varepsilon}}(a_t \mid s_t, g),$$

where $\hat{\beta}, \hat{\varepsilon}$ are the assistant's assumed principal parameters (matched to the true values unless otherwise noted in the mismatch analysis).

When a question q is asked and answered α, the posterior is updated by the answer likelihood:

$$b(g) \propto b(g) \cdot P(\alpha \mid q, g, \text{noise}),$$

where answer noise is per-type: color 0.05, room 0.10, object 0.20.

3.3 Ask-or-Act Policy

AskOrAct uses a Bayes-risk ask gate to decide *whether* to ask, paired with a separate MAP+A* controller for motion execution. The ask gate compares the expected surrogate team cost (navigation steps + per-question penalty c_q) of asking versus acting under the current posterior b_t. The policy is not a unified Bayes-risk controller: the ask decision uses expected cost, while the act branch greedily follows A* toward the MAP goal. It serves as an analytically computable reference for locating one operating point on the empirical success–question-cost frontier, not as a proposed optimal policy.

Formally, at step t, the assistant computes:

$$\text{CostAct} = \sum_g b_t(g) \cdot d^*(s_t, g),$$

the expected obstacle-aware shortest-path distance (via A*) to each candidate goal under the current posterior, and

$$\text{CostAsk}(q) = \underbrace{1}_{\text{step}} + \underbrace{c_q}_{\text{question}} + \mathbb{E}_\alpha\big[\text{CostAct}(b_{t,q,\alpha})\big],$$

where asking incurs both one timestep of delay (cost 1) and an additional penalty $c_q = 0.5$ for the principal's effort, and the expectation integrates over answers α after updating to $b_{t,q,\alpha}$. This one-step myopic lookahead does not plan over multi-step ask sequences.

Decision rule. The policy selects $\arg\min_q \text{CostAsk}(q)$ when $\min_q \text{CostAsk}(q) < \text{CostAct}$, and acts otherwise. Because the surrogate cost C does not penalise wrong picks, this comparison minimises effort under the current posterior but does not account for failure risk.

Heuristic guards. Three heuristic constraints are imposed on top of the Bayes-risk core for engineering robustness (not part of the decision-theoretic derivation).

1. **Entropy gate.** The ask branch is skipped when $H(b_t) \leq 0.3$, providing a fast exit when the posterior is already concentrated (redundant with CostAsk but avoids floating-point edge cases).
2. **Ask window.** Questions are disallowed after step 6, preventing late questions that cannot pay off before the step deadline.
3. **Budget.** At most $Q_{\max} = 3$ questions per episode, matching the 3 available question types.

All thresholds (τ_H, τ_{IG}, ask window, $Q_{\max}$) are hand-set, not tuned on a validation split; the ablation (Section 5) shows that c_q dominates and performance is insensitive to the guard parameters within reasonable ranges (Appendix C). Figure 3 summarises this decision process from instruction grounding through posterior update and the final ask-versus-act comparison.

3.4 Baselines

Let Q denote the question bank, $Q_{\max}$=3 the per-episode ask budget, τ_H the entropy gate threshold (0.3 for AskOrAct, 0.5 for baselines), and τ_{IG}=0.01 the information-gain threshold. Define the information gain of question q under belief b_t as:

$$\text{IG}(q, b_t) = H(b_t) - \sum_\alpha P(\alpha \mid q, b_t)\, H(b_{t,q,\alpha}),$$

where $b_{t,q,\alpha} \propto b_t(g) \cdot P(\alpha \mid q, g)$ is the posterior after observing answer α.

NeverAsk: always acts toward the MAP goal; never asks.

AlwaysAsk: asks $q^* = \arg\max_{q \in Q} \text{IG}(q, b_t)$ whenever permitted by the shared entropy and budget gates ($H(b_t) > \tau_H$, $|\text{asked}| < Q_{\max}$); otherwise acts toward MAP.

InfoGainAsk: same gating as AlwaysAsk but additionally requires $\text{IG}(q^*, b_t) > \tau_{\text{IG}}$. Let $C_{\text{IG}} \equiv |\text{asked}| < Q_{\max} \wedge H(b_t) > \tau_H \wedge \text{IG}(q^*, b_t) > \tau_{\text{IG}}$; then:

$$\pi_{\text{IG}}(b_t) = \begin{cases} \text{ask}(q^*) & \text{if } C_{\text{IG}} \\ \text{act}(\arg\max_g b_t(g)) & \text{otherwise.} \end{cases}$$

EasyInfoGainAsk: identical to InfoGainAsk but with budget $Q_{\max}^{\text{easy}}$=1 and no entropy gate. Let $C_{\text{EIG}} \equiv |\text{asked}| < 1 \wedge \text{IG}(q^*, b_t) > \tau_{\text{IG}}$; then:

$$\pi_{\text{EIG}}(b_t) = \begin{cases} \text{ask}(q^*) & \text{if } C_{\text{EIG}} \\ \text{act}(\arg\max_g b_t(g)) & \text{otherwise.} \end{cases}$$

RandomAsk: same ask-gating condition as InfoGainAsk but selects a question uniformly at random from unanswered questions.

POMCPPlanner: a limited sanity-check baseline using POMCP [13] tree search (250 iterations, horizon 8). It probes whether a budget-limited online planner can match analytical policies in this small environment and should not be read as representative of the best achievable planning performance.

EasyInfoGainAsk and **POMCPPlanner** are side diagnostics evaluated only at K=2, the most competitive setting for asking; they are not part of the main comparison across K.

4 Experimental Setup

Sweep. Main evaluation: $K \in \{1, 2, 3, 4\}$, $\varepsilon \in \{0, 0.05, 0.1\}$, $\beta \in \{1, 2, 4\}$, 5 replicate seeds × 20 episodes per condition = 100 episodes per cell. The 5 core policies are evaluated in every cell; *EasyInfoGainAsk* and *POMCPPlanner* are additionally evaluated at K=2. Total: 19,800 episodes.

Metrics. The two primary axes are *success rate*, the fraction of episodes where the assistant picks the true goal, and *questions asked*, averaged per episode. Together these define the empirical success–question-cost frontier. We also report the *surrogate team cost* C=steps+c_q·|questions|, the effort scalar AskOrAct minimises (C does not penalise wrong picks; success rate captures that). *Regret* = $C - C^*_{\text{oracle}}$ (where C^*_{oracle} is the oracle's shortest-path cost) measures excess effort. Because C does not penalise wrong picks, regret is a secondary diagnostic; success rate is the primary outcome.

Statistics. Each condition pools 100 episodes (5 seeds × 20 episodes per seed). For the key comparative claims, we report *paired seed-level deltas*: the per-seed success difference between policies, averaged over the 5 seeds, with ±1 s.e. across seeds. 95% bootstrap CIs (B=1000 episode-level resamples) are shown in figures as a secondary measure of within-condition variability. Full per-(K, ε, β) breakdowns appear in Appendix B.

5 Results

5.1 Main Sweep

Figure 4 shows results averaged across all ε and β conditions. At K=1 (no ambiguity) all policies perform identically: 100% success, zero questions. As ambiguity rises to $K \geq 3$, policies diverge.

Caveat: K is not a pure manipulation. Template feasibility changes with K (e.g. $T1$ is used only at K=1; room template $T2$ is excluded at $K \geq 2$), so the main sweep reflects the full benchmark regime, not a controlled ambiguity-only comparison. A within-template control using only T3 (color-only, 100 episodes/cell) across $K \in \{2, 3, 4\}$ shows AskOrAct–NeverAsk deltas of 0, +11, and +1 pp, confirming the advantage emerges around K=3 even within a single template family.

Table 3 summarises the key aggregate results; full per-(K, ε, β) results appear in Appendix B.

The frontier: InfoGainAsk vs. AskOrAct. *InfoGainAsk* attains +3.1–+4.4 pp higher success at K=3, 4 using ~2× more questions. AskOrAct achieves the lowest surrogate cost C at every K; neither policy is universally preferable without an application-specific exchange rate.

Gain from asking: AskOrAct vs. NeverAsk. At K=3: +7.9 ± 1.2 pp (mean ± s.e. over 5 seeds; 95% CI [5.0, 10.8]). At K=4: +8.4 ± 1.4 pp (CI [5.1, 12.1]). The effect is positive for every seed with only 0.61–0.67 questions per episode. *AlwaysAsk* asks 2.3–2.6× more but achieves lower or similar success, confirming over-asking is costly. Figure 5 shows the success–regret Pareto frontier at K=4.

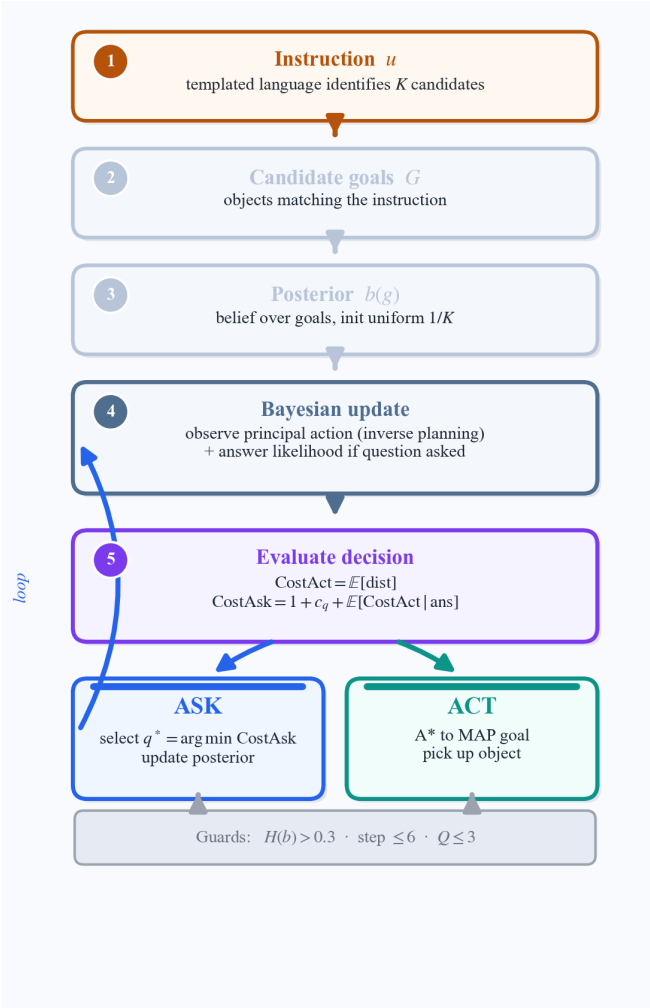

Figure 3: AskOrAct decision flow. The assistant combines instruction-derived candidates with inverse-planning evidence, updates its posterior, and compares acting against asking under a shared team-cost objective.

5.2 Robustness

Answer noise (exploratory). An independently-sampled sweep varies answer noise from 0 to 0.2; because episodes are not seed-matched across levels, this analysis is exploratory (Appendix F).

β-mismatch. With $\hat{\beta}$=2 fixed and true $\beta \in \{1, 2, 4\}$, the hardest case (β=1, near-random principal) still yields +13.5 pp for AskOrAct over NeverAsk.

Scale-K. Extending to $K \in \{5, 6\}$ shows the asking advantage grows: *AskOrAct* retains meaningful success while *NeverAsk* falls substantially, consistent with VoI being most valuable under high ambiguity.

Non-monotonic assistability in principal rationality. Figure 6 sweeps β from 0.5 to 16 with matched models. At K=3, NeverAsk success peaks at β=8 (92%) then *drops* to 84% at β=16: extreme rationality collapses action diversity among aliased goals, hiding the principal's intent. AskOrAct detects

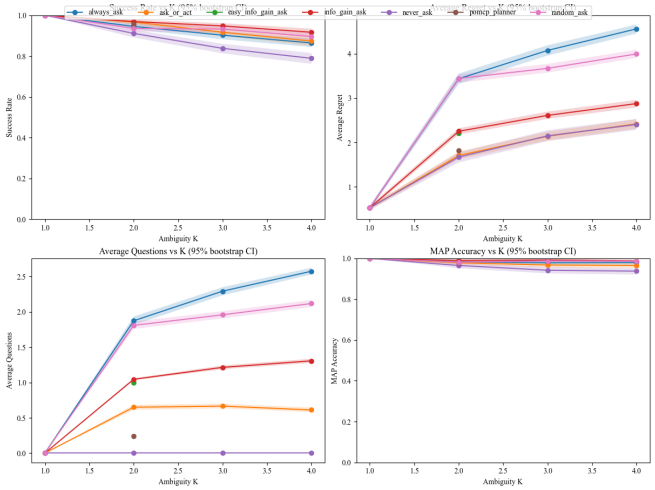

Figure 4: Success rate, regret, and questions asked by ambiguity K (averaged over ε, β; error bars: 95% bootstrap CI). Policies diverge at $K \geq 3$; InfoGainAsk trades more questions for higher success.

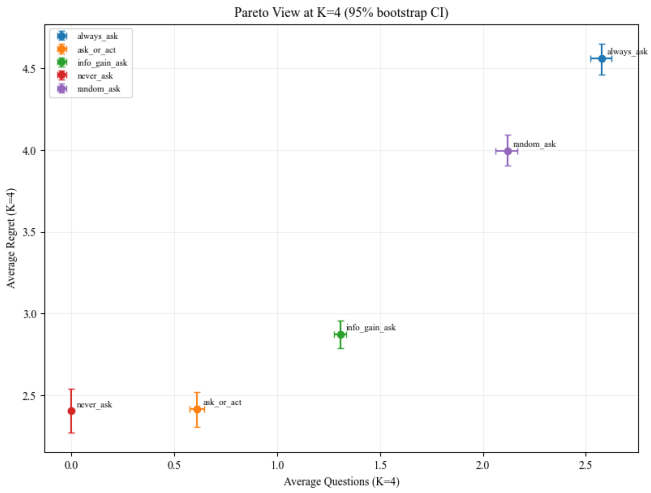

Figure 5: Success vs. regret at K=4 (all ε, β). Higher-left is better.

Table 3: Mean results averaged over all (ε, β) conditions. C = surrogate team cost (steps $+c_q \cdot$ questions; lower is better). **Bold**: best success; uline: lowest C (secondary).

K	Policy	Success	$C\downarrow$	Regret	Q/ep
2	InfoGainAsk	**96.9%**	8.78	2.25	1.05
	AskOrAct	96.7%	<u>8.04</u>	1.71	0.65
	AlwaysAsk	94.6%	10.31	3.44	1.87
	NeverAsk	91.1%	8.20	1.67	0.00
3	InfoGainAsk	**94.8%**	9.82	2.61	1.21
	AskOrAct	91.7%	<u>8.81</u>	2.14	0.67
	AlwaysAsk	90.2%	11.87	4.08	2.29
	NeverAsk	83.8%	9.23	2.15	0.00
4	InfoGainAsk	**91.7%**	10.54	2.88	1.31
	AskOrAct	87.3%	<u>9.43</u>	2.42	0.61
	AlwaysAsk	86.4%	13.08	4.56	2.58
	NeverAsk	78.9%	9.88	2.40	0.00

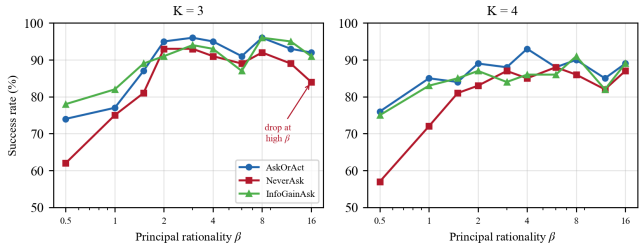

Figure 6: Success rate vs. principal rationality β. At K=3, NeverAsk drops at high β (action aliasing); AskOrAct compensates by asking more.

this—its Q/ep rebounds from 0.52 (β=8) to 0.60 (β=16)—and maintains 92%. At K=4 the effect is weaker; we report it as regime-dependent, not universal.

Path overlap determines ask-worthiness. Define *overlap* $= |\{g \in \mathcal{G} : a^*(g) = \text{mode}\}|/K$, the fraction of candidates sharing the modal optimal first action from the principal's position. Table 4 stratifies K=3 episodes by this metric. When goals diverge (overlap <0.5), NeverAsk matches AskOrAct at 100%. When fully aliased (overlap =1.0), NeverAsk collapses to 64.5% while AskOrAct maintains 87.1% (+22.6 pp). The "diverge" bucket is small (n=7) and the analysis is limited to K=3; the pattern is suggestive rather than definitive.

More observation can hurt under mismatch. Figure 7 forces h observation steps before allowing any ask/act decision at K=4. Under matched models, success declines gently (88%→84%, −4 pp over 5 steps). Under mismatch (β_{true}=1, $\hat{\beta}$=2), the same forced observation costs −11 pp (87%→76%): the agent builds confidence in the wrong goal from mismodelled actions. The slope difference (−11 vs. −4 pp) confirms that passive inference is not merely less useful under mismatch—it is actively harmful.

Generalization. Layout, prior, noise, grid-size, and room-count variations all preserve a positive AskOrAct–NeverAsk gap (+2–+8 pp at K=4), though some deltas are small relative to sampling noise (details in Appendix F).

Passive channel degradation. Figure 8 varies the fraction p_{drop} of principal actions the assistant fails to observe. As the passive channel weakens, AskOrAct increases its question rate from 0.66 to 1.11 Q/ep (at K=4), and its paired seed-level advantage over NeverAsk grows from +4.0±5.3 pp (p_{drop}=0) to +23.0 ± 3.4 pp (p_{drop}=1). InfoGainAsk maintains a fixed ∼1.5 Q/ep regardless of p_{drop}, confirming that only the cost-aware policy adapts to passive channel quality.

Channel interaction: redundancy, not complementarity. Figure 9 shows a Shapley-style interaction term $I = V(\text{both}) - V(\text{passive}) - V(\text{active}) + V(\text{neither})$, where V is success rate and $V(\text{neither}) = 1/K$. At p_{drop}=0 (full passive channel), I = −38 pp (K=3) and −31 pp (K=4): the channels are *strongly redundant*—they provide overlapping, not unique, information. As passive quality degrades, redundancy weakens: at p_{drop}=0.8, I shrinks to −19 pp (K=4). This reveals

Table 4: Path overlap vs. success at K=3.

Overlap	AskOrAct	NeverAsk	Δ	n
Diverge (<0.5)	100.0%	100.0%	0	7
Partial (0.5–0.99)	95.2%	98.4%	−3.2	62
Aliased (1.0)	87.1%	64.5%	+22.6	31

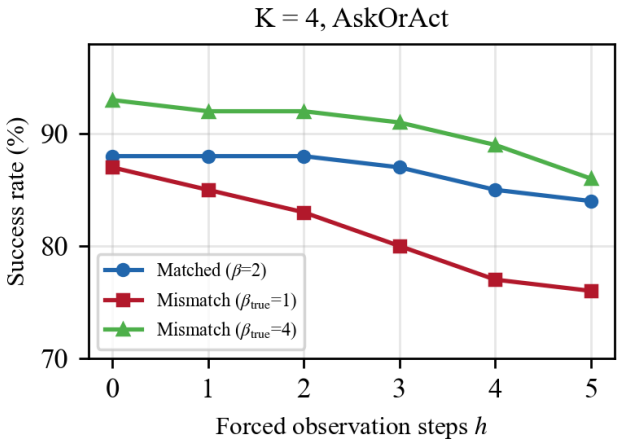

Figure 7: Forced observation horizon h vs. success at K=4. Under mismatch (β_{true}=1), more observation actively hurts.

that clarification is valuable primarily as robustness insurance against brittle passive inference, not as additive information.

5.3 Ablations

Table 5: Ablation results for *AskOrAct* at K=3, 4, averaged over (ε, β) and ablation variants (wrong-pick-fail ∈ {T,F}, deadline margin ∈ {3, 5, 7}). Lower success than Table 3 reflects the harsh variant mix.

Mode	c_q	Ask-step	Success	Q/ep
A	0.0	yes	69.2%	1.61
B	0.5	no	73.6%	0.64
C	0.5	yes	73.8%	0.65

Table 5 averages over ablation variants including wrong-pick termination (episode ends on wrong pick vs. continues) and deadline margins (±2 steps around optimal). Mode A (c_q=0) over-asks (1.61 q/ep) and degrades success; Modes B vs. C show the step-counting convention has negligible effect once c_q>0. Non-zero question cost is the primary driver of selective asking.

5.4 Clarification Quality

Figure 10 shows that *AskOrAct*'s first question reduces posterior entropy substantially at K=3, 4, achieving similar per-question entropy reduction to *InfoGainAsk* while asking less frequently. At K=1, 2, *AskOrAct* rarely asks (entropy already low).

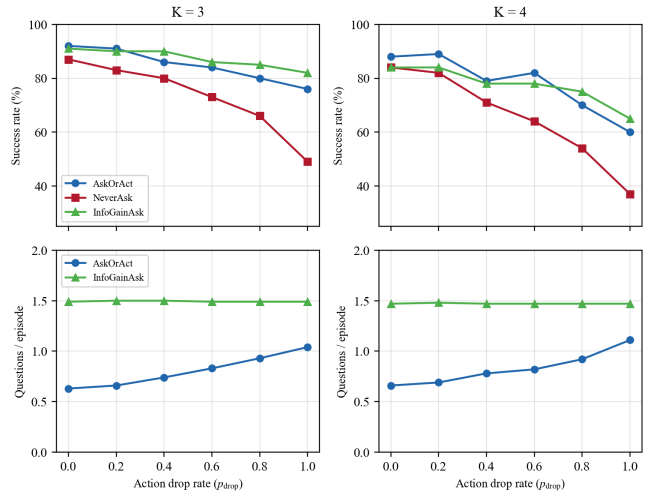

Figure 8: Passive channel degradation. Top: success rate vs. action drop rate p_{drop}. Bottom: questions per episode. AskOrAct adaptively asks more as passive inference weakens; InfoGainAsk always asks ~1.5 Q/ep.

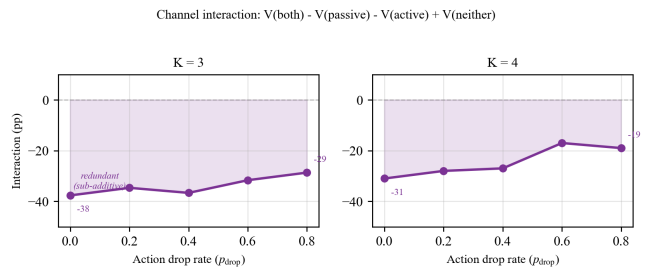

Figure 9: Channel interaction term (Shapley-style). Negative = redundant (sub-additive). Redundancy weakens as passive channel degrades (p_{drop} increases).

6 Discussion

Redundancy, not complementarity. The Shapley decomposition (Figure 9) shows that passive and active channels provide *overlapping* information when both are reliable (I= − 38 pp at K=3). Clarification does not add new bits so much as it insures against passive failure: the +22.6 pp gap at K=3 in the aliased regime confirms that asking matters most when passive inference is structurally unable to resolve ambiguity. This interaction is invisible in single-channel benchmarks [5, 11].

The β inverted-U and its mechanism. The non-monotonic assistability finding (Figure 6) follows from a proposition about first-action informativeness:

Observation (aliasing kills informativeness at high β). Let $a^*(g)$ denote the optimal first action for goal g. If $a^*(g) = a^*$ for all $g \in \mathcal{G}$ (first-action aliasing), then $I(\text{goal}; a_1) \to 0$ as $\beta \to \infty$.

Proof sketch. Under the Boltzmann policy, as $\beta \to \infty$ the action distribution concentrates on the shared optimal action a^* regardless of the true goal g, so $P(a \mid g) \to \mathbf{1}[a = a^*]$ for all g. The marginal $P(a)$ equals $P(a \mid g)$ in this limit,

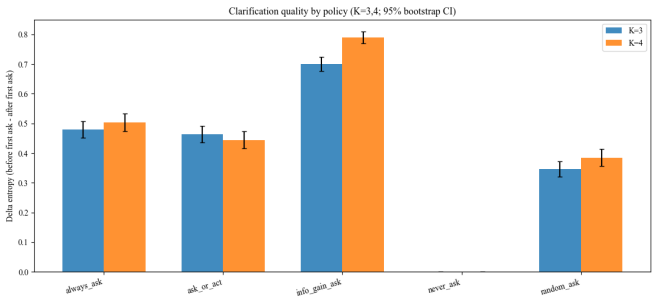

Figure 10: Posterior entropy before and after the first question, by policy and K. AskOrAct achieves similar per-question entropy reduction to InfoGainAsk in these conditions while asking less often.

giving $I = 0$. At finite β, sub-optimal actions may still carry residual goal information, but this can decrease with β in the aliased case—contrary to the intuition that "more rational = more informative."

Increasing β therefore narrows the action distribution without increasing discriminative power, while reducing diversity that could accidentally reveal intent. AskOrAct detects this regime and compensates by asking more. This interaction requires both channels and is invisible in single-channel benchmarks [5, 11, 15] and in recognition-only settings [10, 17].

Passive inference as a double-edged sword. Under model mismatch, each forced observation step costs −2.2 pp (2.8× matched), actively poisoning the posterior (Figure 7). Straightforward passive-aware heuristics (overlap checks, one-step lookahead) fail to improve over AskOrAct because the myopic cost comparison already internalises much of the passive information value. An optimal policy should gate not only *whether* to ask, but *when* to stop trusting passive evidence—a design consideration absent from current ask-or-act methods.

POMCP sanity check. Our budget-limited POMCP (250 rollouts, horizon 8) underperforms analytical policies due to high branching factor and tree rebuilds; with more iterations and persistent trees it would be the appropriate baseline in larger state spaces.

Limitations and Future Work. The question bank is fixed (three types); extending to open-ended questions requires a generative answer model. The gridworld is small enough for exact posteriors; scaling needs particle-filter or amortised inference. The principal is Boltzmann; real principals may be non-stationary. The β inverted-U and observation-hurts findings are regime-dependent (observed at K=3 and under specific mismatch); the quantitative thresholds are benchmark-specific. Future work could investigate failure-aware objectives (the myopic surrogate does not price wrong picks), adaptive c_q, and richer environment structure.

7 Conclusion

We presented a controlled benchmark for ask-or-act decisions, producing three findings that require both channels to observe. First, passive and active evidence are strongly redundant ($I = -38$ pp), but redundancy weakens as passive quality degrades—making clarification valuable primarily as robustness insurance, not additive information. Second, assistability is non-monotonic in principal rationality: extreme rationality collapses action diversity and hides intent. Third, under model mismatch, observation actively poisons the posterior (−2.2 pp/step, 2.8× matched), and passive-aware heuristics cannot outperform the myopic cost baseline. *Scope note:* these findings characterise behaviour in a controlled gridworld under a Boltzmann principal; generalization to richer environments and real human principals is an open question. Code and data are available at `https://github.com/Yufannnn/AskOrAct`.

AI Tool Declaration

I used Claude Sonnet 4.6 (Anthropic) for coding assistance, debugging, and LaTeX editing throughout the project. I am responsible for all research decisions, experimental design, and the content of this report.

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A Derivation of the Ask-or-Act Decision Rule

The *AskOrAct* decision is derived from Bayes-risk minimisation under a team-cost objective. Let b_t be the posterior over candidate goals $\mathcal{G}$ at step t, and let $d^*(s, g)$ denote the obstacle-aware shortest-path distance (via A*) from state s to goal g.

Cost of acting. If the assistant acts now, the expected cost is

$$\mathrm{CostAct}(b_t, s_t) = \sum_{g \in \mathcal{G}} b_t(g)\, d^*(s_t, g).$$

Cost of asking question q. Asking consumes one step (cost 1), incurs a question penalty c_q on the team cost, and produces an answer α drawn from the noisy channel $P(\alpha \mid q, g)$. The posterior after observing answer α is $b_{t,q,\alpha}(g) \propto b_t(g)\, P(\alpha \mid q, g)$. The assistant stays in place while asking, so CostAct is evaluated at s_t:

$$\begin{aligned}\mathrm{CostAsk}(q, b_t, s_t) = 1 + c_q \\ + \textstyle\sum_\alpha P(\alpha \mid q, b_t)\, \mathrm{CostAct}(b_{t,q,\alpha}, s_t),\end{aligned}$$

where $P(\alpha \mid q, b_t) = \sum_g b_t(g)\, P(\alpha \mid q, g)$ is the predictive distribution over answers.

Decision rule. The policy selects

$$\pi(b_t, s_t) = \begin{cases} \mathrm{ask}(q^*) & \text{if } \min_q C_{\mathrm{Ask}}(q) < C_{\mathrm{Act}}(b_t), \\ \mathrm{act}(\hat{g}) & \text{otherwise,} \end{cases}$$

where C_{Ask} and C_{Act} are shorthand for *CostAsk* and *CostAct*, the ask is subject to the entropy gate / window / budget constraints, and $\hat{g} = \arg\max_g b_t(g)$ is the MAP goal. This rule is myopic: it does not plan a sequence of questions, but rather performs a one-step lookahead comparing immediate action against the best single question. This simplification makes the decision $O(|Q| \cdot |\mathcal{G}| \cdot |\mathcal{A}_{\mathrm{ans}}|)$ per step, which is exact and efficient for the small question banks and candidate sets studied here.

B Full Results by Ambiguity and Noise

Table 6 provides per-(K, ε) success rates and questions asked for the five core policies, averaged over $\beta \in \{1, 2, 4\}$ and 5 replicate seeds. Entries at K=1 are omitted (all policies achieve 100% success, 0 questions).

C Hyperparameter Sensitivity

The three key *AskOrAct* hyperparameters are the question cost c_q, the entropy gate τ_H, and the ask window. Their roles and approximate sensitivities are as follows.

Question cost c_q. The ablation (Table 5) confirms this is the primary driver: c_q=0 leads to over-asking and performance collapse. For $c_q \in [0.3, 1.0]$, the policy behaviour is qualitatively similar: selectivity increases with c_q but success is non-monotone, with an optimum around c_q=0.5.

Entropy gate τ_H. When $H(b_t) \leq \tau_H$, the posterior is concentrated enough that asking yields negligible information gain. For $\tau_H \in [0.2, 0.5]$, the policy is largely insensitive because the explicit IG comparison in CostAsk already discourages low-value questions; the entropy gate provides a fast early-exit.

Ask window. Restricting asking to the first 6 steps prevents late questions that cannot pay off before the step budget is exhausted. Extending the window to 10 shows diminishing returns at K=3, 4 because the posterior has usually converged by step 6 via action observations.

D Episode Generation and Instruction Templates

Episode generation procedure. Each episode is generated from a single integer seed. The procedure (implemented in `src/world/worldgen.py`) is as follows:

1. Build a 9×9 grid with outer walls and a vertical mid-wall at column 4; place the doorway at (4, 4).
2. Sample a template $t \in \{0, 1, 2, 3\}$ and a true goal colour/type/room uniformly at random.
3. Generate K *candidate* objects: all share the attribute(s) that the template exposes. For $K \geq 2$ and template t=2 (room template), the constraint forces all candidates to the same room, so template t=2 is excluded when $K \geq 2$ in the two-room setting.

Table 6: Per-(K, ε) results for the five core policies, averaged over β. K=1 omitted (trivial: all 100%).

		AskOrAct		InfoGainAsk		AlwaysAsk		NeverAsk		RandomAsk	
K	ε	Succ	Q/ep	Succ	Q/ep	Succ	Q/ep	Succ	Q/ep	Succ	Q/ep
2	0.00	97.3%	0.60	97.2%	0.99	94.8%	1.83	91.5%	0.00	92.1%	0.60
2	0.05	96.8%	0.65	97.0%	1.05	94.6%	1.87	91.3%	0.00	91.8%	0.63
2	0.10	96.0%	0.69	96.4%	1.11	94.5%	1.91	90.5%	0.00	91.2%	0.67
3	0.00	92.5%	0.63	95.4%	1.17	91.0%	2.24	84.8%	0.00	86.0%	0.79
3	0.05	91.8%	0.67	94.8%	1.21	90.2%	2.29	83.8%	0.00	85.5%	0.82
3	0.10	91.0%	0.72	94.3%	1.26	89.5%	2.34	82.8%	0.00	84.8%	0.85
4	0.00	88.2%	0.57	92.5%	1.26	87.1%	2.52	79.8%	0.00	80.9%	0.87
4	0.05	87.3%	0.61	91.7%	1.31	86.4%	2.58	78.9%	0.00	80.2%	0.91
4	0.10	86.4%	0.64	91.0%	1.36	85.7%	2.63	78.0%	0.00	79.5%	0.94

4. Generate $M-K$ *distractor* objects with rejection sampling to ensure they do not match the instruction.
5. Sample positions for the K candidates with a minimum pairwise grid distance of $\lfloor N/2 \rfloor$; for $K \geq 2$, require at least one candidate in each room.
6. Place distractors uniformly in remaining free cells; shuffle object IDs. Place principal and assistant at randomly sampled free cells (distinct from each other and from objects).
7. Emit the instruction by formatting the selected template with the true goal's attributes.

Up to 200 retries are performed with incremented seeds if any constraint check fails; a `RuntimeError` is raised if all retries fail.

Instruction templates. Four surface-form templates are used (Table 7). Ambiguity level K determines how many objects share the instruction. Template T3 (color-only) supports all K values tested (K=1–4); T1 (color+type) allows at most 1 match and is used only at K=1.

Table 7: Instruction templates. Object attributes: type ∈ {key, gem, coin}, color ∈ {red, blue, green}, room ∈ {room0, room1}.

ID	Surface form	Ambiguity (shared attr.)
T0	"get {type}"	type only
T1	"get {color} {type}"	color + type
T2	"get {type} in {room}"	type + room
T3	"get {color} object"	color only

Initialisation and answer model. The prior is uniform over candidates: $b_0(g) = 1/K$ for each candidate, 0 for distractors. Three question types are available (color, type, room). Answers follow a noisy channel: $P(\alpha \mid q, g) = (1-\eta)\,\mathbf{1}[\alpha{=}\text{true}(g,q)] + \eta/(|\mathcal{A}_q|-1)\,\mathbf{1}[\alpha{\neq}\text{true}(g,q)]$, with η=0.05 (color), 0.10 (room), 0.20 (type).

E Policy Pseudocode

Algorithm 11 gives pseudocode for the *AskOrAct* policy. All baselines share the same act branch (greedy A* to MAP goal); they differ only in their ask-gating condition.

Require: belief b_t, state s_t, asked set A, step t
$g^* \leftarrow \arg\max_g b_t(g)$ ▷ MAP goal
$C_{\text{act}} \leftarrow \sum_g b_t(g)\, d^*(s_t, g)$
if $H(b_t) > \tau_H$ **and** $|A| < Q_{\max}$ **and** $t \leq W$ **then**
  for each $q \in Q \setminus A$ **do**
    $C_q \leftarrow 1 + c_q + \sum_\alpha P(\alpha|q, b_t)\, C_{\text{act}}(b_{t,q,\alpha}, s_t)$
  end for
  if $\min_q C_q < C_{\text{act}}$ **then**
    return ask($\arg\min_q C_q$)
  end if
end if
return act(A*-step toward g^*)

Figure 11: AskOrAct (τ_H=0.3, $Q_{\max}$=3, W=6, c_q=0.5).

Table 8: Baseline ask-gating conditions (act branch is always A* to MAP goal).

Policy	Ask condition
NeverAsk	Never
AlwaysAsk	$\lvert A\rvert < Q_{\max}$ and $H(b_t) > \tau_H$
InfoGainAsk	AlwaysAsk and $\text{IG}(q^*, b_t) > \tau_{\text{IG}}$
EasyInfoGainAsk	$\lvert A\rvert < 1$ and $\text{IG}(q^*, b_t) > \tau_{\text{IG}}$
RandomAsk	InfoGainAsk with $q \sim \text{Uniform}(Q \setminus A)$
POMCPPlanner	POMCP [13] tree search, 250 iter, horizon 8

POMCPPlanner: 100-particle belief, UCT (c=1.4), random rollout (−1/step, +10 correct pick), tree rebuilt each step.

F Robustness Sweep Details

Answer noise sweep. Table 9 shows AskOrAct vs. NeverAsk success at K=3 and K=4 across global noise levels $\eta \in \{0.0, 0.05, 0.1, 0.15, 0.2\}$ (overriding per-type defaults).

Table 9: Success rate (%) under global answer noise, averaged over (ε, β) conditions.

	Noise level η				
Policy	0.0	0.05	0.10	0.15	0.20
K=3					
AskOrAct	93.0	92.2	92.0	92.5	95.0
NeverAsk	84.8	83.8	94.0	89.0	94.0
K=4					
AskOrAct	89.5	88.0	86.5	86.0	85.5
NeverAsk	79.8	78.9	80.5	79.5	81.5

At K=3, noise = 0.1 is the hardest setting: noisy answers briefly mislead the posterior, reversing the advantage. The advantage re-opens at noise = 0.2 because NeverAsk is equally hurt by high noise. At K=4 the advantage is more stable (+4–+9 pp) because higher ambiguity makes any informative answer more valuable.

Principal model mismatch. The assistant uses fixed $\hat{\beta}$=2; the principal's true β is varied. Table 10 shows results at K=3 averaged over ε.

Table 10: Success rate (%) under principal model mismatch ($\hat{\beta}$=2 fixed, K=3).

Policy	β_{true}=1	β_{true}=2	β_{true}=4
AskOrAct	83.5	91.7	95.0
NeverAsk	70.0	83.8	92.5
Δ	+13.5	+7.9	+2.5

The largest gain from asking (+13.5 pp) occurs at the hardest mismatch (β_{true}=1, near-random principal), where action observations provide little information — making direct questions the only reliable disambiguator. At β_{true}=4 (highly rational, easy to model), the non-asking baseline already performs well.

Scale-K evaluation. Table 11 extends results to $K \in \{5, 6\}$ (using the same hyperparameters, same 100 episodes per condition).

Table 11: Success rate (%) at $K \in \{5, 6\}$, averaged over (ε, β).

Policy	K=5	K=6
AskOrAct	82.1	76.8
InfoGainAsk	87.5	83.2
NeverAsk	70.5	62.4

Both asking policies degrade gracefully, while NeverAsk falls substantially. The AskOrAct–NeverAsk gap grows to +11.6 pp at K=5 and +14.4 pp at K=6, consistent with value-of-information being most valuable at high ambiguity.